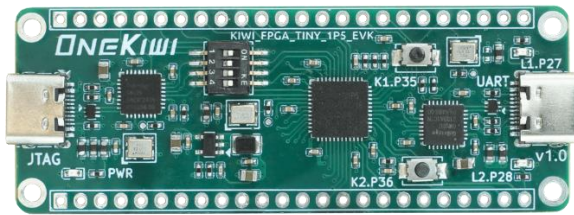


KIWI 1P5 FPGA BOARD

Gowin FPGA 1K5 LUTs, onboard high-speed USB-UART converter



- Main FPGA Chip: GW1N-UV1P5 on chip Flash
- Embedded 1584 FPGA LUTs
- Onboard Gowin GWU2X USB-JTAG debugger
- Onboard Button and LED
- **On-chip Flash – no need external Flash IC**
- Support JTAG DIP Switch program external FPGA

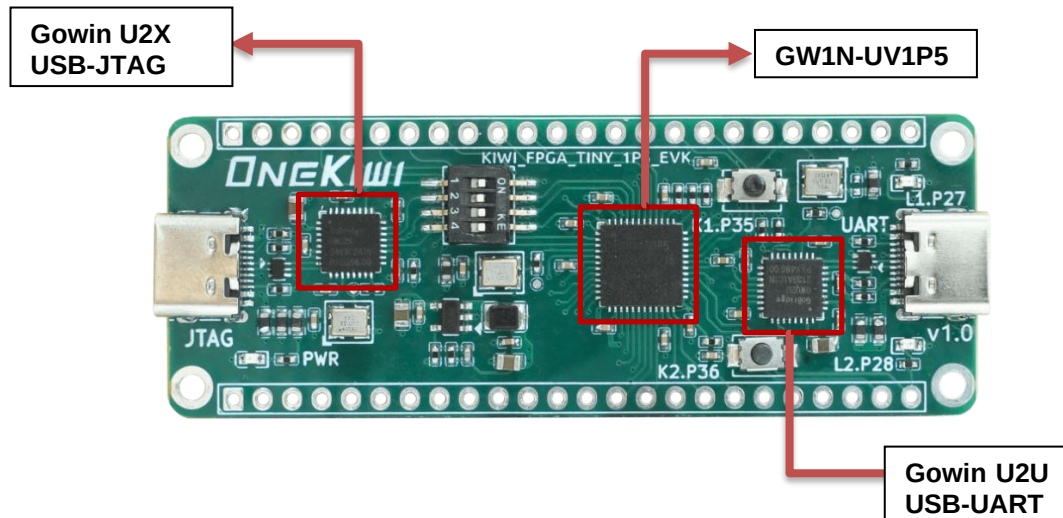
Kiwi 1P5		
FPGA Details	FPGA Chip	GW1N-UV1P5
	Logic Units	1584
	Hard processor	None
	Block SRAM (bits)	96K
	User Flash (bits)	256K
	Number of PLL	2
	Number of I/O Bank	6
	Number of user I/O	125
User Interface	User Button	02
	User LED	02
Programmer	Programmer IC	Gowin U2X Support Programming for External FPGA (Change Switch SW1)
Physical	Size	70mm x 22 mm
	Interface	Double row header pitch 2.54mm
Power	Main Voltage	5V
	I/O Voltage	All IO 3.3V
Environment	Temperature	Industrial: -40°C to 75°C
	Humidity	Humidity (Non-condensing): 10% - 90%

KIWI 1P5 FPGA BOARD

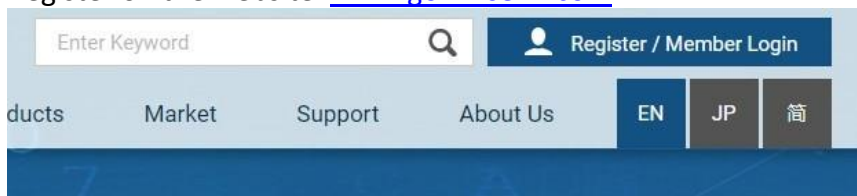
Gowin FPGA 1K5 LUTs, onboard high-speed USB-UART converter

Quick Start Guide

- Kiwi 1P5 Peripheral Description



- Register on the website: www.gowinsemi.com



- Download the GOWIN EDA Development Environment and Apply License
https://www.gowinsemi.com/en/support/download_eda/



Home > Support > GOWIN EDA Home > Apply License

The screenshot shows the Gowin EDA Apply License page with the title "Apply License". It includes a sidebar with navigation links and a main form area with the following fields:

- * Compulsory field
- Company Name *
- Company Website

Block Diagram and Pinout Table

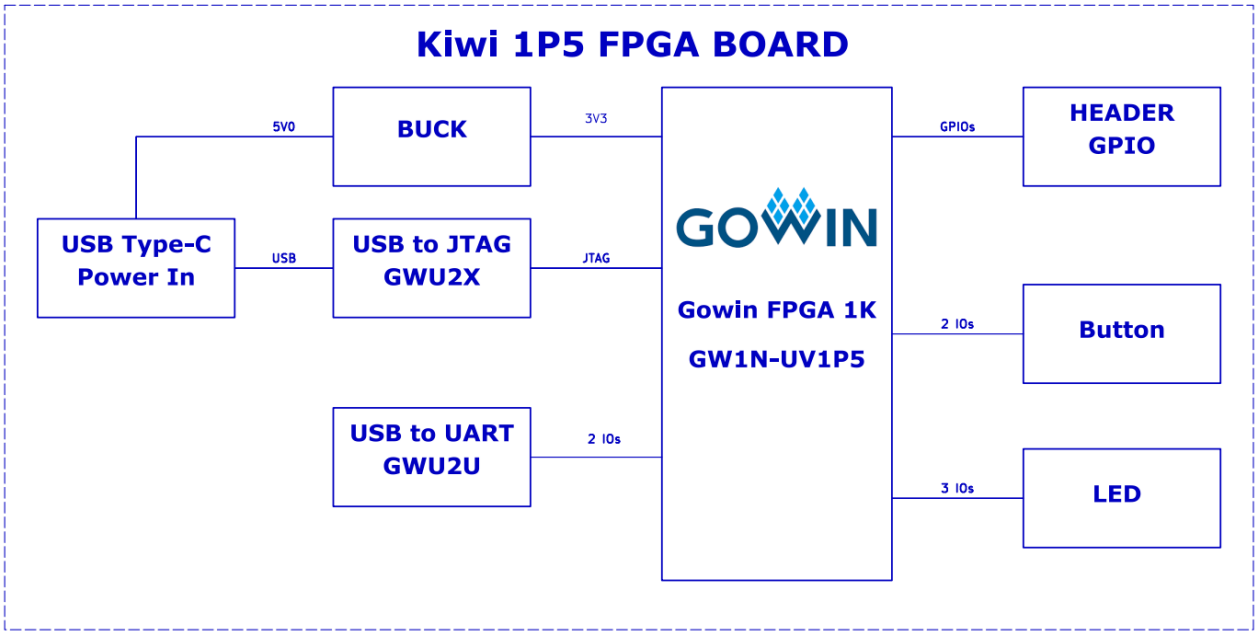


Fig 01. Kiwi 1P5 Block Diagram

KIWI 1P5 FPGA BOARD

Gowin FPGA 1K5 LUTs, onboard high-speed USB-UART converter

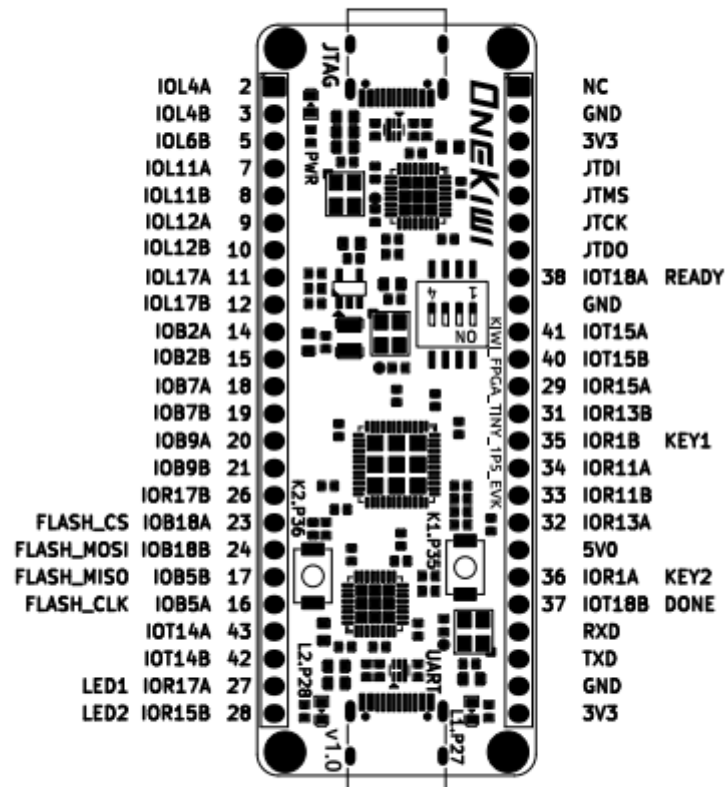


Fig 02. Kiwi 1P5 Pinout Mapping (All IO 3.3V voltage level)

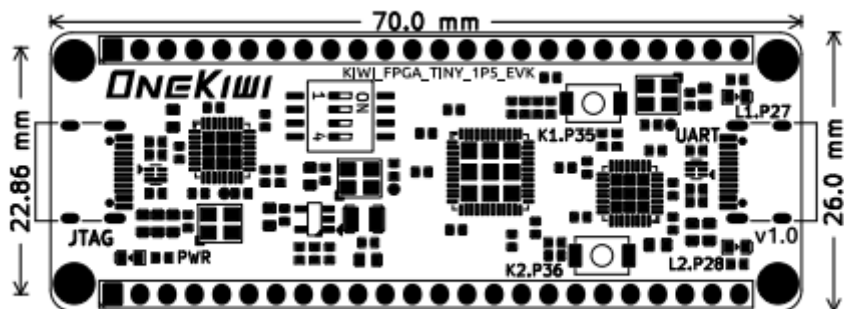
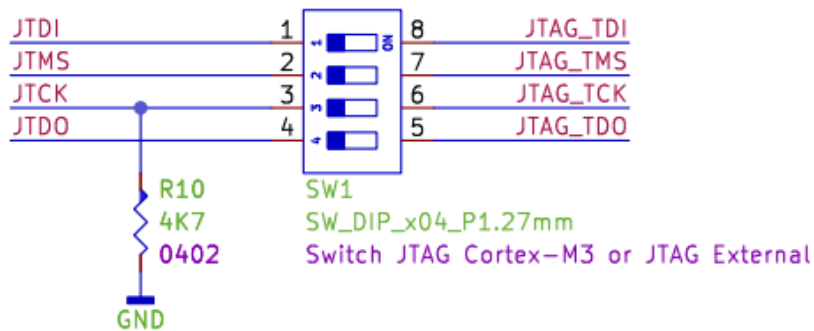


Fig 03. Kiwi 1P5 board dimension

KIWI 1P5 FPGA BOARD

Gowin FPGA 1K5 LUTs, onboard high-speed USB-UART converter

- JTAG Switch function instruction

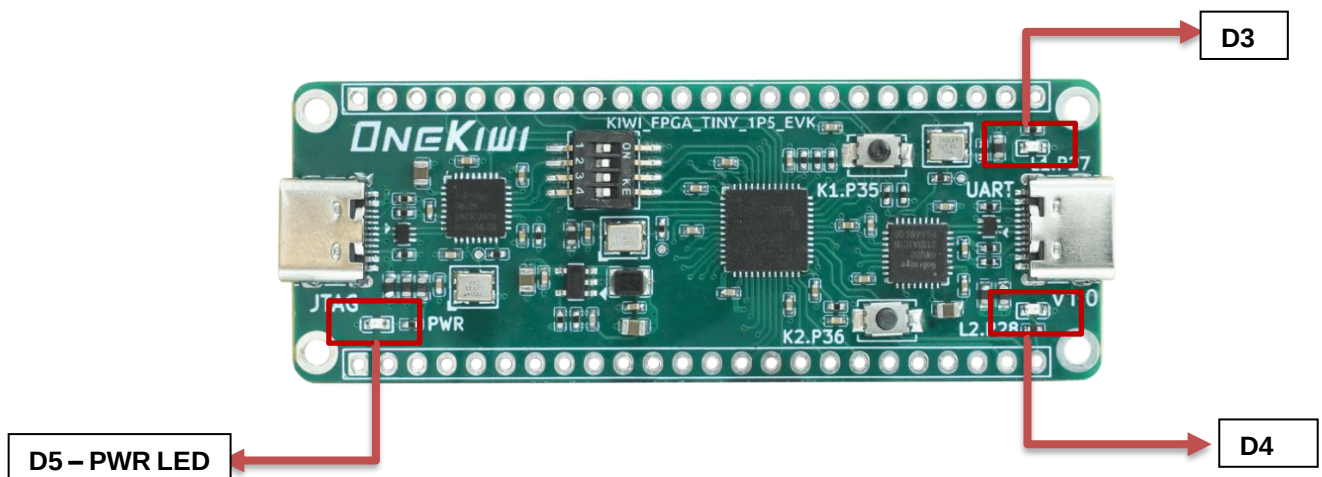
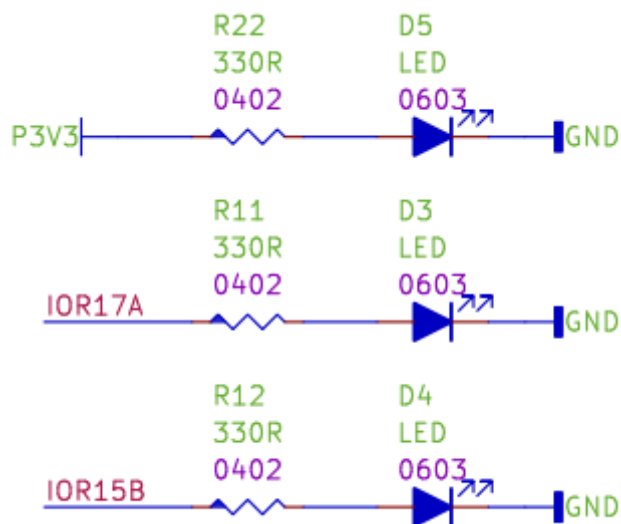


Switch ON: USB JTAG connected to FPGA chip

Switch OFF: Disconnect FPGA chip from USB-JTAG, can use programmer with external FPGA chip

- LED circuit description

There are total 3 LEDs available on the 1P5 board D3, D4, D5, maximum voltage 3.3V.

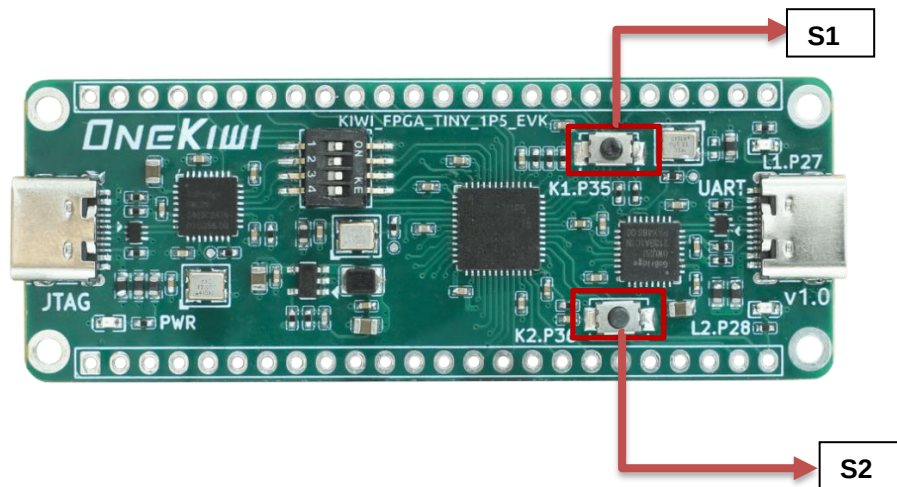
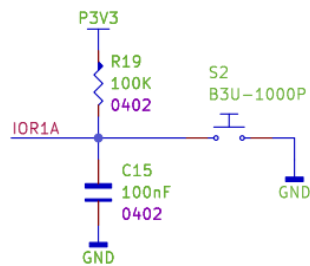
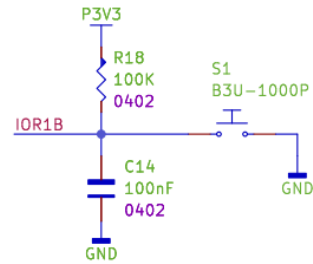


KIWI 1P5 FPGA BOARD

Gowin FPGA 1K5 LUTs, onboard high-speed USB-UART converter

- **Button circuit description**

There are total 2 User buttons default Active Low with voltage level 3.3V



KIWI 1P5 FPGA BOARD

Gowin FPGA 1K5 LUTs, onboard high-speed USB-UART converter

- FPGA Pin assign list table

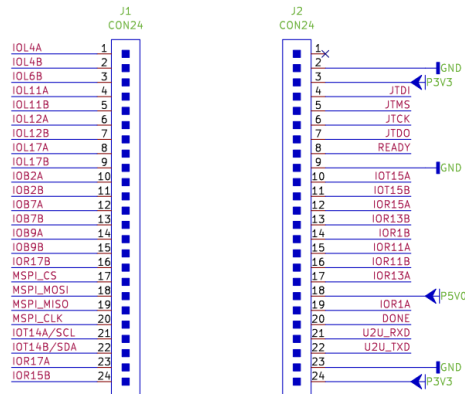


Table 01: Pinout description CON24 – J1

Signal Name	Pin No.	J1 Pin No.	BANK	Description	I/O
IOL4A	2	1	5	General I/O	VCCIO5
IOL4B	3	2	5	General I/O	VCCIO5
IOL6B	5	3	5	General I/O	VCCIO5
				Global Clock	
IOL11A	7	4	4	General I/O	VCCIO4
IOL11B	8	5	4	General I/O	VCCIO4
IOL12A	9	6	4	General I/O	VCCIO4
				Global Clock	
IOL12B	10	7	4	General I/O	VCCIO4
IOL17A	11	8	3	General I/O	VCCIO3
IOL17B	12	9	3	General I/O	VCCIO3
IOB2A	14	10	2	General I/O	VCCIO2
IOB2B	15	11	2	General I/O	VCCIO2
IOB7A	18	12	2	General I/O	VCCIO2
IOB7B	19	13	2	General I/O	VCCIO2
				Global Clock	
IOB9A	20	14	2	General I/O	VCCIO2
				Global Clock	
IOB9B	21	15	2	General I/O	VCCIO2
				Global Clock	
IOR17B	26	16	2	General I/O	VCCIO2
MSPI_CS	23	17	2	General I/O (Remapping)	VCCIO2
				Special Pin MSPI	
MSPI_MOSI	24	18	2	General I/O (Remapping)	VCCIO2
				Special Pin SPI SI	
MSPI_MISO	17	19	2	General I/O (Remapping)	VCCIO2
				Special Pin SPI SO	
MSPI_CLK	26	20	2	General I/O (Remapping)	VCCIO2
				Special Pin SPI CLK	
IOT14A/SCL	43	21	0	General I/O (Remapping)	VCCIO0
				Special Pin I2C CLK	
IOT14B/SDA	42	22	0	General I/O (Remapping)	VCCIO0
				Special Pin I2C DATA	
IOR17A	27	23	2	General I/O	VCCIO2
IOR15B	28	24	2	General I/O	VCCIO2

KIWI 1P5 FPGA BOARD

Gowin FPGA 1K5 LUTs, onboard high-speed USB-UART converter

Table 02: Pinout description CON24 – J2

Signal Name	Pin No.	J2 Pin No.	BANK	Description	I/O
NC	-	1	-	General I/O	
GND	-	2	-	General I/O	
P3V3	-	3	-	General I/O	
JTDI	-	4	-	General I/O	
JTMS	-	5	-	General I/O	
JTCK	-	6	-	General I/O	
JTDO	-	7	-	General I/O	
READY	38	8	0	General I/O JTAG READY	VCCIO0
GND	-	9	-	General I/O	
IOT15A	41	10	0	General I/O	VCCIO0
IOT15B	40	11	0	General I/O	VCCIO0
IOR15A	29	12	2	General I/O	VCCIO2
IOR13B	31	13	2	General I/O	VCCIO2
IOR1B	35	14	2	General I/O	VCCIO2
IOR11A	34	15	2	General I/O Global Clock	VCCIO2
IOR11B	33	16	2	General I/O Global Clock	VCCIO2
IOR13A	32	17	2	General I/O	VCCIO2
P5V0	-	18	-		
IOR1A	36	19	2	General I/O	VCCIO2
DONE	37	20	0	JTAG DONE General I/O (Remapping)	VCCIO0
U2U_RXD	-	21	-	U2U Rx Pin	
U2U_TXD	-	22	-	U2U Tx Pin	
GND	-	23	-	General I/O	
P3V3	-	24	-	General I/O	

KIWI 1P5 FPGA BOARD

Gowin FPGA 1K5 LUTs, onboard high-speed USB-UART converter

Development Board Suite

A development board suite includes the following items:

- 1 x KIWI 1P5 development board (1)
- 1 x USB Type-C cable (2)
- 2 x Header 2.54mm 24 pin (3)
- Carton Box (4)

